

Assignment for Week 1

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```
library(dplyr, warn.conflicts = FALSE) library(ggplot2) library(nycflights13) library(MASS) knitr::opts_chunk$set(echo = TRUE)
```

```
set.seed(100)
```

Assignment Part I

3.13 Let A, B, C be subsets of a set X . Which of the expressions $(A \cap B) \cup (A \cap B^c)$, $(A \cap B) \cup (A \cap B^c)^c$, $(A \cap B) \cup (A^c \cup C)^c$ can be simplified, and how?

Answer: all the expressions can be simplified as follows:

$$(A \cap B) \cup (A \cap B^c) = A,$$

$$(A \cap B) \cup (A \cap B^c)^c = A^c \cup B,$$

$$(A \cap B) \cup (A^c \cup C)^c = A \cap (B \cup C^c).$$

3.4.2 Let g be the mapping from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined by

$$g(x, y) = (y, -x),$$

and let the mapping h be as in Exercise 3.4.1. Find the image of (x, y) under each of the composite mappings $g \circ h$ and $h \circ g$.

Answer: h is defined by

$$h(x, y) = (x, -y),$$

and so

$$(g \circ h)(x, y) = g(x, -y) = (-y, -x),$$

and

$$(h \circ g)(x, y) = h(y, -x) = (y, x).$$

Therefore, the image of (x, y) under $g \circ h$ is $(-y, -x)$, and the image of (x, y) under $h \circ g$ is (y, x) .

31.1.1 As in the text, let $\mathbb{Q}$ be the set of all rational numbers. If $x \in \mathbb{Q}$, which of the following statements (P, Q, R, S) imply which of the others?

P: $x \geq 0$

Q: $\exists w \in \mathbb{Q}$ such that $x = w^2$

R: $nx > -1 \ \forall n \in \mathbb{N}$

S: $\exists n \in \mathbb{N}$ such that $nx > -1$.

Answer: P, Q, and R are all equivalent to $x \geq 0$, and S is true for all $\mathbb{Q}$. Hence we have the following implications:

$$P \Rightarrow Q, P \Rightarrow R, P \Rightarrow S,$$

$$Q \Rightarrow P, Q \Rightarrow R, P \Rightarrow S,$$

$$R \Rightarrow P, R \Rightarrow Q, R \Rightarrow S.$$

5.1.3 In each of the following cases, state whether the sequence $\{u_n\}$ tends to a limit, and find the limit if it exists:

(a) $u_n = 1 + \frac{1}{2}n$

(b) $u_n = 1 - \frac{1}{2}n$

(c) $u_n = (\frac{1}{2})^n$

(d) $u_n = -(\frac{1}{2})^n$

Answer:

(a) The limit does not exist.

(b) The limit does not exist.

(c) $\lim_{n \rightarrow \infty} u_n = 0$

(d) $\lim_{n \rightarrow \infty} u_n = 0$

Assignment Part II

Package Loading

```
library(dplyr, warn.conflicts = FALSE)
library(ggplot2)
library(nycflights13)
```

(1) What are the types of data that serve as the argument and value of the following functions?

- `nchar()`
- `log()`
- `paste()`
- `is.na()`

Answer:

The argument of `nchar()` is a character vector, and the value of `nchar()` is a vector.

The argument of `log()` is any positive number, and the value of `log()` is any real number.

The argument of `paste()` is one or more R object, and the value of `paste()` is a character vector.

The argument of `is.na()` is an R object to be tested, and the value of `is.na()` is a logical vector.

(2) What was the longest physical distance taken out of New York in 2013?

Code: `max(flights$distance)`

[1] 4983

Answer: the longest physical distance taken out of New York in 2013 is 4983 miles.

(3) In R, we can use scientific notation. $2e3$ is 2000. Why is $\log(1e10) - \log(1e8)$ the same as $\log(1e2)$?

Answer: this is because $\log(1e10) - \log(1e8) = \log((1e10)/(1e8)) = \log(1e2)$.

(4) Without running these commands, what are the values of C, D, and E?

```
A <- "Iowa"
B <- "Kansas"

C <- !((A == "Kentucky") | (B == "Kansas"))

D <- ((nchar(A) > 2) & is.na(B)) | (!is.na(A))

E <- ifelse( C, D, A)
```

Answer: the value of C is FALSE; the value of D is FALSE; the value of E is "Iowa"

(5) Plot an interesting subset of the data Flights data. Use the Rmarkdown `fig.height` and `fig.width` to make it look nice.

Answer: The subset of the data I choose is flights from New York to Boston (BOS) on April 15, 2013, the day on which the Boston Marathon bombings occurred. I want to compare the delay of flights on that day with delay on other days in April, 2013 from New York to Boston.

The graph is on page 5. The preliminary conclusion from the plot is that there is not significant delay of flights from New York to Boston due to the Marathon bombings. It is worth checking if flights departing from, instead of arriving at, Boston on that day were significantly delayed.

Code:

```
flightscomparison <- mutate(flights, month == 4, Marathonday = ifelse(day == 15, "Marathon Day",
"Non-Marathon Day"))
ggplot(flightscomparison, aes(dep_time, group = Marathonday, fill = Marathonday)) + geom_density(alpha=0.5)
```

(6) Write your own “quantile” function which gets the median and the 95th percentile of data.

Answer:

```
# Create the vector.
```

```
x <- c(7,3,4.2,18,2,-21,8)
```

```
# Find the 95th percentile.
```

```
quantile(x, c(0.025, 0.05, 0.25, 0.5, 0.75, 0.95, 0.975)) # Find the median.
```

```
median.result <- median(x)
```

(7) What does `df` look like when it is printed out?

```
df <- data.frame(dogheight=c("10", "20", "30"), dogwidth=c("5", "3", "1"))
df[df$dogwidth==3,1] <- NA
```

Answer: `df` is a dataframe of 2 variables, namely `dogheight` and `dogwidth`, and 3 observations. The `dogheight` corresponding to a `dogwidth` of 3 is not available.

(8) *True or False*

- (a) A small p-value, such as .001 or .0001, means that our estimate is unlikely to have arisen from the null hypothesis.
- (b) The small errors distribution is a nice approximation for the distances traveled by flights out of New York City.
- (c) $f(x) = 2x$ for $0 \leq x \leq 1$ is a possible probability density function.
- (d) The probability that a normal distribution produces the number 0 is 0.

Answer: (a) *False* (b) *False* (c) *True* (d) *False*

